PRATICAL-6

AIM:- Perform the logistic regression on the given data warehouse data using R/Python.

STEPS:-

To perform this you need to download quality.csv file from following link:
https://github.com/TarekDib03/Analytics/tree/master/Week3%20- %20Logistic%20Regression/Data

Step 1:- provide path of file where it is saved on your machine

```
quality <- read.csv('C:/Users/Gauri/Downloads/quality.csv')

> #analysing the quality dataset

> str(quality)

'data.frame': 131 obs. of 14 variables:

$ MemberID : int 1 2 3 4 5 6 7 8 9 10 ...

$ InpatientDays : int 0 1 0 0 8 2 16 2 2 4 ...

$ ERVisits : int 0 1 0 1 2 0 1 0 1 2 ...

 $ OfficeVisits : int 18 6 5 19 19 9 8 8 4 0 ...

$ Narcotics : int 1 1 3 0 3 2 1 0 3 2 ...

$ DaysSinceLastERVisit: num 731 411 731 158 449 ...

 $ Pain : int 10 0 10 34 10 6 4 5 5 2 ...

$ TotalVisits : int 18 8 5 20 29 11 25 10 7 6 ...

$ ProviderCount : int 21 27 16 14 24 40 19 11 28 21 ...

$ MedicalClaims : int 93 19 27 59 51 53 40 28 20 17 ...

 $ ClaimLines : int 222 115 148 242 204 156 261 87 98 66 ...

$ StartedOnCombination: logi FALSE FALSE FALSE FALSE FALSE FALSE ...

 $ AcuteDrugGapSmall : int 0 1 5 0 0 4 0 0 0 0 ...

$ PoorCare : int 0 0 0 0 0 1 0 0 1 0 ...

 > table(quality$PoorCare)

0 1

98 33

> 98/131

[1] 0.7480916
```

> install.packages("caTools")

Installing package into 'C:/Users/Saurav/Documents/R/win-library/3.5' (as 'lib' is unspecified)

--- Please select a CRAN mirror for use in this session ---

Step 2:- installing the dependency 'bitops' from

'http://mirror.its.dal.ca/cran/bin/windows/contrib/3.5/bitops_1.0-6.zip' Content type 'application/zip' length 38894 bytes (37 KB)

installing the dependency 'caTools' from

'http://mirror.its.dal.ca/cran/bin/windows/contrib/3.5/caTools_1.17.1.1.zip'

Step 3:- package 'bitops' successfully unpacked and MD5 sums checked

package 'caTools' successfully unpacked and MD5 sums checked

The downloaded binary packages are in C:\Users\Saurav\AppData\Local\Temp\RtmpmUN9oK\downloaded_package s

Step 4:- > library(caTools) Warning message: package 'caTools' was built under R version 3.5.2

> set.seed(88)

> split = sample.split(quality$PoorCare, SplitRatio = 0.75)

>

> split

[1] TRUE TRUE TRUE TRUE FALSE TRUE FALSE TRUE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE

[28] TRUE TRUE FALSE FALSE FALSE FALSE TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE FALSE TRUE TRUE FALSE TRUE FALSE TRUE FALSE TRUE TRUE FALSE FALSE TRUE

[55] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE

[82] TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE

[109] TRUE FALSE FALSE TRUE TRUE FALSE TRUE TRUE TRUE FALSE TRUE TRUE FALSE TRUE TRUE FALSE TRUE TRUE TRUE FALSE

> qualityTrain = subset(quality, split == TRUE)

> qualityTest = subset(quality, split == FALSE) > nrow(qualityTrain)

[1] 99

> nrow(qualityTest)

[1] 32

> QualityLog = glm(PoorCare ~ OfficeVisits + Narcotics,data=qualityTrain, family=binomial)

```
> summary(QualityLog)
Call:
glm(formula = PoorCare ~ OfficeVisits + Narcotics, family = binomial, data = qualityTrain)
 Deviance Residuals:
Min 1Q Median 3Q Max
 -2.06303 -0.63155 -0.50503 -0.09689 2.16686
Coefficients:
Estimate Std. Error z value Pr(>|z|)
 (Intercept) -2.64613 0.52357 -5.054 4.33e-07 ***
OfficeVisits 0.08212 0.03055 2.688 0.00718 **
Narcotics 0.07630 0.03205 2.381 0.01728 *
 -Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
 (Dispersion parameter for binomial family taken to be 1)
 Null deviance: 111.888 on 98 degrees of freedom
Residual deviance: 89.127 on 96 degrees of freedom
AIC: 95.127
 Number of Fisher Scoring iterations: 4
 > predictTrain = predict(QualityLog, type="response")
> summary(predictTrain)
 Min. 1st Qu. Median Mean 3rd Qu. Max.
0.06623 0.11912 0.15967 0.25253 0.26765 0.98456
> tapply(predictTrain, qualityTrain$PoorCare, mean)
0          1
 0.1894512         0.4392246
> table(qualityTrain$PoorCare, predictTrain > 0.5)
 FALSE TRUE
0 70 4
1 15 10
 > 10/25
[1] 0.4
> 70/74
```

[1] 0.9459459

> table(qualityTrain$PoorCare, predictTrain > 0.7)

FALSE TRUE

0 73 1

1 17 8

> 8/25

[1] 0.32

> 73/74

[1] 0.9864865

> table(qualityTrain$PoorCare, predictTrain> 0.2)

FALSE TRUE

0 54 20

1 9 16

> 16/25

[1] 0.64

> 54/74

[1] 0.7297297

> install.packages("ROCR")

Installing package into 'C:/Users/Saurav/Documents/R/win-library/3.5' (as 'lib' is unspecified) also installing the dependencies 'gtools', 'gdata', 'gplots'

trying URL

'http://mirror.its.dal.ca/cran/bin/windows/contrib/3.5/gtools_3.8.1.zip'

'http://mirror.its.dal.ca/cran/bin/windows/contrib/3.5/gplots_3.0.1.1.zip'

'http://mirror.its.dal.ca/cran/bin/windows/contrib/3.5/ROCR_1.0-7.zip'

> library(ROCR)

Loading required package: gplots

Attaching package: 'gplots'

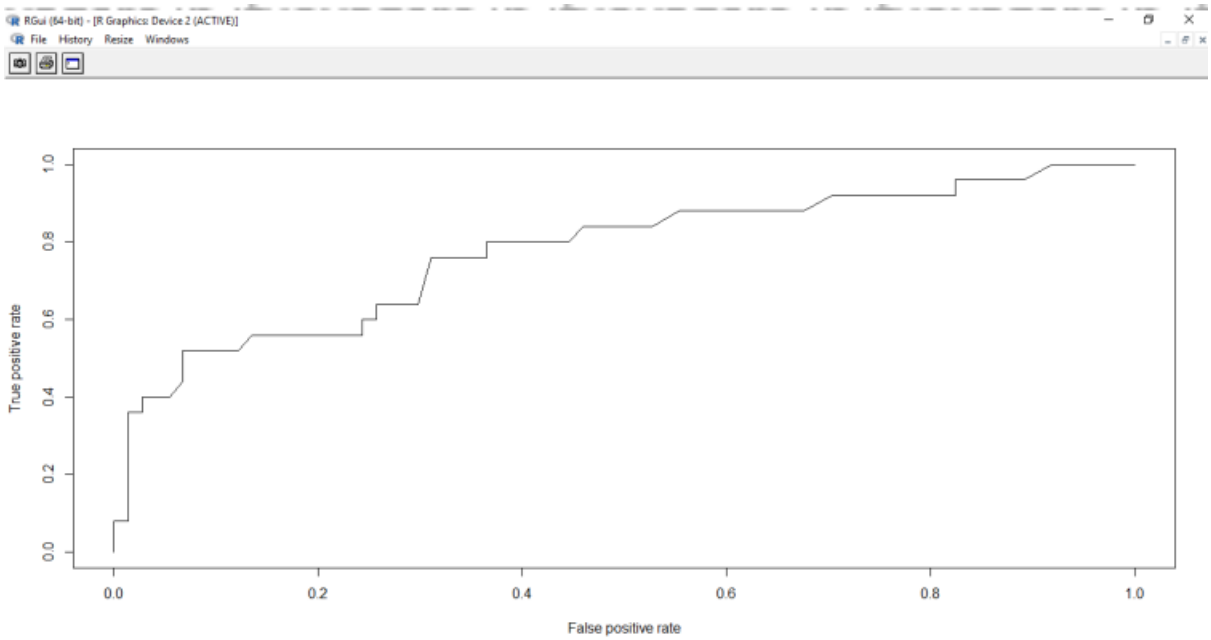
The following object is masked from 'package:stats':

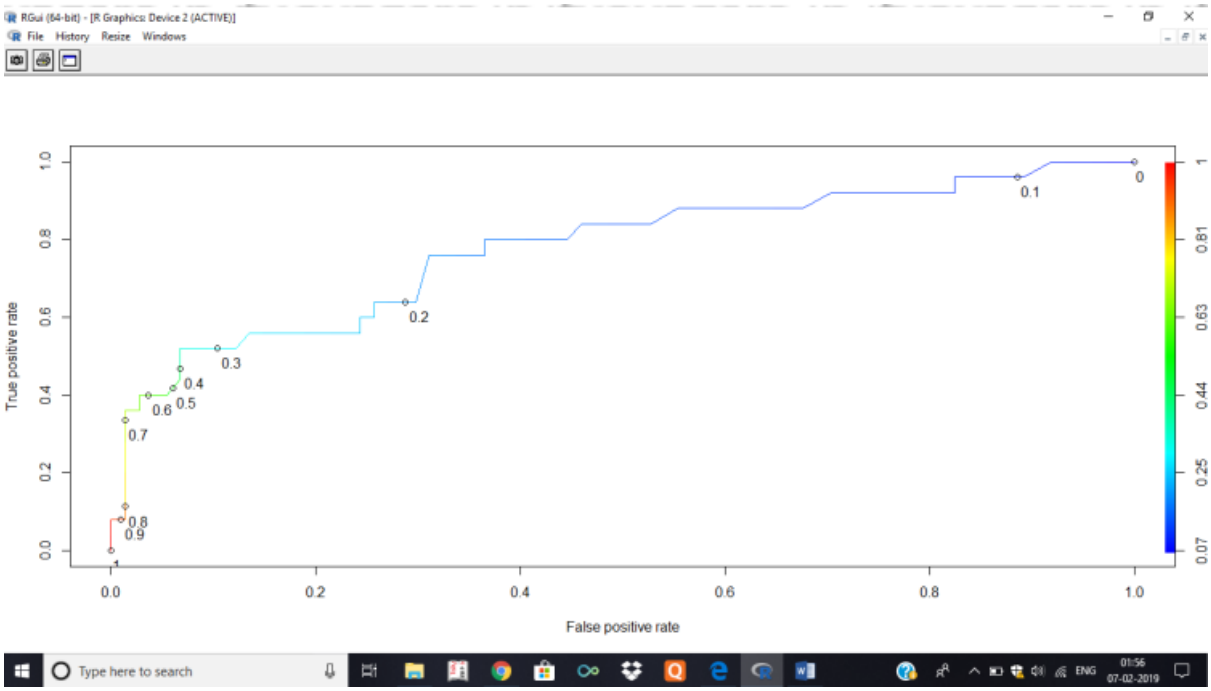
Step 5:- Warning messages:

1: package 'ROCR' was built under R version 3.5.2

2: package 'gplots' was built under R version 3.5.2

```
> ROCRpred = prediction(predictTrain, qualityTrain$PoorCare)
> ROCRperf = performance(ROCRpred, "tpr", "fpr")
> plot(ROCRperf)
> plot(ROCRperf, colorize=TRUE)
> plot(ROCRperf, colorize=TRUE, print.cutoffs.at=seq(0,1,by=0.1), text.adj=c(-0.2,1.7))
>
```




RGui (64-bit) - [R Graphics: Device 2 (ACTIVE)]
File History Resize Windows
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
0.07
0.25
0.44
0.63
0.81
1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Type here to search
ENG
01:56
07-02-2019